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# Hunter Ellis

## Autonomous Systems Engineer

ellishw.tech  
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### Skills

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**Languages:** C++, C, MATLAB, Lua,  $\LaTeX$

**Software:** Git, ROS2, OMPL, Qt, Simulink, FreeRTOS, Autodesk Inventor (Certified), SolidWorks, Rhino

**Hardware:** PCB Design, Breadboarding, Computer Architecture, Oscilloscope, Multimeter, 3D-Printing

### Technical Experience

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#### Autonomous Systems Engineer

Aug 2025 – Present

Georgia Tech Research Institute · Applied Systems Lab

Atlanta, GA

- Software development for collaborative autonomous UAV swarms.
- Motion planning research for swarm path planning
- Collaborative swarm autonomy
- Software systems integration
- embedded software development
- C++ development
- DDS

#### Robotics Research / M.S. Computer Engineering

Aug 2023 – May 2025

Virginia Tech · Control, Optimization, and Online Learning for Autonomy Lab

Blacksburg, VA

- Built a 6-DOF robotic arm and an accompanying ROS2-Gazebo simulation environment for training, evaluating, and deploying custom control algorithms.
- Integrated object detection (YOLOv8), natural language processing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks.
- Aided professors in teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design.

#### Thrust Vector Control Intern

May 2024 – Aug 2024

Jacobs Space Exploration Group · Thrust Vector Control Test Lab

Huntsville, AL

- Developed thrust vector control testing hardware and software for NASA's Active Inertial Load Simulator at the Marshall Space Flight Center.
- Created and ran tests to develop a mathematical model of an electro-mechanical actuator – used Python, MATLAB, and LabVIEW.

#### Control Research Intern

Jun 2023 – Aug 2023

G2ELab · Power Electronics Lab

Grenoble, France

- Researched different inverter control methods for 4-leg (microgrid) topologies in Simulink and HIL simulation.

#### Naval Research Enterprise Internship Program

Jun 2022 – Aug 2022

NSWC Carderock · Center for Innovation in Ship Design

Bethesda, MD

- Developed concept hospital sea-train designs and estimated fuel consumption and electrical power loads of the concept sea-trains.

### Projects

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#### 6-DOF Robot Arm

- 3D printed robot arm, built using stepper motors and pulleys.
- Implemented ROS2 Jazzy control and Gazebo Harmonic simulation.

### Education

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#### M.Sc. in Computer Engineering

May 2025

Virginia Tech · Bradley Department of Electrical & Computer Engineering

Blacksburg, VA

- Focus: Software and Machine Intelligence
- Research: UT Austin's Control, Optimization, and Online Learning for Autonomy Lab

#### B.Sc. in Electrical & Computer Engineering (Double Major)

May 2024

Virginia Tech · Bradley Department of Electrical & Computer Engineering

Blacksburg, VA

- Focus: Controls, Robotics, and Autonomy
- Design Teams: SolarCar at VT and Human Powered Submarine